Lecture 10. Simulation

Inverse transform, rejection method, discrete distributions, variance reduction

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10.1 Random numbers and simulation

Intractable probabilities and expectations are estimated by **simulation**: average many independent realizations (strong law). Everything is built from **random numbers**, i.i.d. $U(0,1)$ (pseudorandom in practice); e.g. $\lfloor nU \rfloor + 1$ is uniform on $\{1, \dots, n\}$.

Example 10.1 (A random permutation)

Start from any ordering $P_1, \dots, P_n$; for $k = n, \dots, 2$ choose I uniform on $\{1, \dots, k\}$ and swap P_I, P_k. Position n gets each element w.p. $1/n$, then position $n-1$ each remaining one w.p. $1/(n-1)$, and so on: probability $1/n!$. The first r positions filled give a random r-subset (random assignment to treatments).

10.2 Simulating continuous random variables

Motivation. All distributions are built from uniforms, by inverse transforms or rejection.

10.2.1 The inverse transformation method

Proposition 10.1

If F is a continuous cdf and $U \sim U(0,1)$, then $X = F^{-1}(U)$ has cdf F.

Proof. $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, since F is increasing. ■

Example 10.2

(a) Exponential: $X = -\frac{1}{\lambda}\log(1-U)$, or $-\frac{1}{\lambda}\log U$.
(b) Gamma(n, λ), integer n: $X = -\frac{1}{\lambda}\log(U_1 U_2 \cdots U_n)$ (one logarithm).

10.2.2 The rejection method

Suppose we can simulate from g and $f(y)/g(y) \le c$ for all y.

Theorem 10.2 (Rejection method)

Repeat: generate $Y \sim g$ and an independent $U \sim U(0,1)$, until $U \le f(Y)/\{cg(Y)\}$; then set $X = Y$. The resulting X has density f, and the number of iterations is geometric with mean c.

Proof. In one iteration, $P(\text{accept}) = \int P(U \le \frac{f(y)}{cg(y)}) g(y)\,dy = \int \frac{f(y)}{c}\,dy = \frac{1}{c}$, and

$$P(Y \le x \mid \text{accept}) = c \int_{-\infty}^{x} \frac{f(y)}{cg(y)} g(y)\,dy = F(x).$$

Iterations are independent, so their number is geometric with parameter $1/c$. ■

Example 10.3 (Normal from exponentials)

$|Z|$ has density $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$, $x > 0$. With $g(x) = e^{-x}$,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2} \le \sqrt{\frac{2e}{\pi}} = c \approx 1.32,$$

attained at $x = 1$, and $\frac{f(x)}{cg(x)} = \exp\{-(x-1)^2/2\}$. Generate $Y = -\log U_1$, accept if $U_2 \le e^{-(Y-1)^2/2}$, then attach a random sign; about 1.32 iterations per normal.

Example 10.4 (The polar method)

By Example 6.14, $R\cos\Theta$, $R\sin\Theta$ are i.i.d. $N(0,1)$ for independent $R^2 \sim \mathrm{Exp}(1/2)$, $\Theta \sim U(0, 2\pi)$. Avoiding trigonometry: generate $V_i = 2U_i - 1$ until $S = V_1^2 + V_2^2 \le 1$; then S is uniform and independent of the angle, with $\cos\Theta = V_1/\sqrt{S}$, $\sin\Theta = V_2/\sqrt{S}$. With $R^2 = -2\log S$,

$$X = V_1\sqrt{\frac{-2\log S}{S}}, \qquad Y = V_2\sqrt{\frac{-2\log S}{S}}$$

are i.i.d. $N(0,1)$; the acceptance probability is $\pi/4$.

χ^2_n: $\sum Z_i^2$, or $-2\log(U_1 \cdots U_{n/2})$ for even n, plus one Z^2 for odd n.

10.3 Simulating from discrete distributions

For $P(X = x_j) = p_j$, set $X = x_j$ if $\sum_{i<j} p_i \le U < \sum_{i\le j} p_i$ (discrete inverse transform).

Example 10.5

(a) Geometric: $X = j$ iff $1 - q^{j-1} \le U < 1 - q^j$, i.e. $X = 1 + \lfloor \log U / \log q \rfloor$ (with $1 - U$ for U).
(b) Binomial: $X = \sum_{i=1}^n \mathbb{1}\{U_i < p\}$.
(c) Poisson(λ): counting events of a rate-λ process in $[0,1]$ with gaps $-\log U_i/\lambda$,

$$N = \max\Big\{n : \prod_{i=1}^{n} U_i \ge e^{-\lambda}\Big\}$$

(multiply random numbers until the product drops below $e^{-\lambda}$; subtract 1).

10.4 Variance reduction

Averaging n copies to estimate $\theta = E[X]$ has error $\sqrt{\mathrm{Var}(X)/n}$; reducing the variance is as good as increasing n.

10.4.1 Antithetic variables

For identically distributed X_1, X_2, $\mathrm{Var}(\frac{X_1+X_2}{2}) = \frac{1}{4}\{2\,\mathrm{Var}(X_1) + 2\,\mathrm{Cov}(X_1, X_2)\}$, smaller than for independent copies when $\mathrm{Cov} < 0$. For $X_1 = h(U_1, \ldots, U_m)$ with h monotone in each argument, $X_2 = h(1 - U_1, \ldots, 1 - U_m)$ has the same law and $\mathrm{Cov}(X_1, X_2) \le 0$ (Exercise 10.A for $m = 1$).

10.4.2 Variance reduction by conditioning

By Proposition 7.6, $\mathrm{Var}(E[X \mid Y]) \le \mathrm{Var}(X)$ with the same mean: if computable, average $E[X \mid Y]$ instead of X.

Example 10.6 (Estimating π)

V_1, V_2 i.i.d. $U(-1,1)$, $I = \mathbb{1}\{V_1^2 + V_2^2 \le 1\}$: $E[I] = \pi/4$, $\mathrm{Var}(I) \approx 0.1685$. Conditioning on V_1,

$$E[I \mid V_1] = P(V_2^2 \le 1 - V_1^2 \mid V_1) = \sqrt{1 - V_1^2},$$

so average $\sqrt{1-U^2}$: $\mathrm{Var} = \frac{2}{3} - \frac{\pi^2}{16} \approx 0.0498$, a 3.4-fold reduction. Antithetic pairs $\frac{1}{2}\{\sqrt{1-U^2} + \sqrt{1-(1-U)^2}\}$ do better: about 0.0069 per pair, against 0.0249 for two independent values.

10.4.3 Control variates

If Y is simulated alongside X with known μ_Y, $X + c(Y - \mu_Y)$ has mean θ and variance $\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y) + 2c\,\mathrm{Cov}(X,Y)$, minimized by

$$c^* = -\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X + c^*(Y - \mu_Y)) = \mathrm{Var}(X)(1 - \rho(X,Y)^2).$$

In practice c^* is estimated from the pairs. In Example 10.6, $Y = U^2$ ($E[Y] = 1/3$) controls $\sqrt{1-U^2}$.

10.5 Selected exercises

Motivation. Ross's Chapter 10 problems (book numbering) verify the recipes; Exercise 10.A justifies antithetic negative correlation.

Exercise 10.1 (Random permutations by insertion)

Set $P(1) = 1$. For $k = 2, \dots, n$, generate a random number U, let $J = \lfloor kU \rfloor + 1$, and set $P(k) = P(J)$ and then $P(J) = k$ (so for $J = k$ simply $P(k) = k$). Show that after step k, $(P(1), \dots, P(k))$ is a uniformly random permutation of $1, \dots, k$.

Solution.

Step 1. Step k puts k at position J and moves the old $P(J)$ to position k; J is uniform and independent of earlier steps.

Step 2. The map $\Phi : (\sigma, j) \mapsto \tau$ (σ a permutation of $1, \dots, k-1$, $j \le k$) is a bijection onto permutations of $1, \dots, k$: j is the position of k in τ, and σ is recovered by putting $\tau(k)$ back at j ($j < k$) and deleting position k.

Step 3. (Induction.) If each σ has probability $1/(k-1)!$, each pair (σ, j) has $\frac{1}{(k-1)!} \cdot \frac{1}{k}$, so each τ has $1/k!$. ■

Exercise 10.7 (The maximum of random numbers)

Let $F(x) = x^n$ for $0 < x < 1$.
(a) Simulate from F using a single random number.

(b) Show that $\max(U_1, \dots, U_n)$ has distribution F for independent random numbers $U_1, \dots, U_n$, and so give a second method.

Solution.

(a) $F^{-1}(u) = u^{1/n}$, so $X = U^{1/n}$ (Proposition 10.1): $P(U^{1/n} \le x) = P(U \le x^n) = x^n$.

(b) $P(\max_i U_i \le x) = \prod_i P(U_i \le x) = x^n$: a second (costlier) method. ■

Exercise 10.10 (Choosing the exponential envelope)

In Example 10.3, use the envelope $g_\lambda(x) = \lambda e^{-\lambda x}$, $x > 0$, instead of e^{-x}. Show that the mean number of iterations of the rejection method is minimized when $\lambda = 1$.

Solution.

Step 1. With $f(x) = \sqrt{2/\pi}\, e^{-x^2/2}$,

$$\frac{f(x)}{g_\lambda(x)} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda x - x^2/2} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda^2/2} e^{-(x-\lambda)^2/2},$$

maximal at $x = \lambda$, so the best constant is $c(\lambda) = \sqrt{2/\pi}\, e^{\lambda^2/2}/\lambda$.

Step 2. The mean number of iterations is $c(\lambda)$ (Theorem 10.2), and $\frac{d}{d\lambda} \log c(\lambda) = \lambda - \frac{1}{\lambda}$ changes sign at $\lambda = 1$: minimum $c(1) = \sqrt{2e/\pi} \approx 1.32$. ■

Exercise 10.13 (Polar coordinates of a uniform point in the disc)

Let (X, Y) be uniform on the unit disc, with density $1/\pi$ there, and let (R, Θ) be its polar coordinates, $\Theta \in (0, 2\pi)$.
(a) Show that R and Θ are independent, $R^2 \sim U(0,1)$ and $\Theta \sim U(0, 2\pi)$.
(b) Show that if V_1, V_2 are independent $U(-1, 1)$, then given $V_1^2 + V_2^2 \le 1$ the point (V_1, V_2) is uniform on the unit disc. (Together, (a) and (b) justify the polar method, Example 10.4.)

Solution.

(a)

Step 1. $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$ maps $(0,1) \times (0, 2\pi)$ one-to-one onto the disc minus a null segment, with Jacobian r; by Theorem 6.3

$$f_{R,\Theta}(r, \theta) = \frac{1}{\pi} \cdot r = 2r \cdot \frac{1}{2\pi}, \qquad 0 < r < 1,\ 0 < \theta < 2\pi.$$

Step 2. It factors (Proposition 6.1), and $P(R^2 \le s) = \int_0^{\sqrt{s}} 2r\, dr = s$.

(b) (V_1, V_2) is uniform on $[-1, 1]^2$ (area 4), the disc D has area π; for $A \subseteq D$

$$P((V_1, V_2) \in A \mid (V_1, V_2) \in D) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

■

Exercise 10.A (Antithetic variables are negatively correlated)

Let h be nondecreasing on $(0,1)$ with $E[h(U)^2] < \infty$ for $U \sim U(0,1)$. Show that $\text{Cov}(h(U), h(1-U)) \le 0$.

Solution. [Supplement]

Step 1. $k(u) = h(1-u)$ is nonincreasing, so for all u, v

$$\big(h(u) - h(v)\big)\big(k(u) - k(v)\big) \le 0.$$

Step 2. With $V \sim U(0,1)$ independent of U, take expectations and expand (U, V i.i.d.):

$$0 \ge E[h(U)k(U)] - E[h(U)]E[k(V)] - E[h(V)]E[k(U)] + E[h(V)k(V)] = 2\,\mathrm{Cov}\big(h(U), k(U)\big).$$

(finite by Cauchy–Schwarz, as $k(U) \overset{d}{=} h(U)$).

Step 3. So the antithetic estimator has variance at most $\frac{1}{2}\,\mathrm{Var}(h(U))$, that of two independent copies. ■